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SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4: Implement and evaluate network protocols and services using simulation tools to analyze performance in real-world scenarios. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Comparative Analysis

Code of Conduct:

I Rishiyesh K.S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.S. Rishiyesh

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CO4 ASSESSMENT TOOL 3

CONCEPT MAPPING / COMPARATIVE ANALYSIS

Course: COMPUTER NETWORKS | Code: CSA07

Details

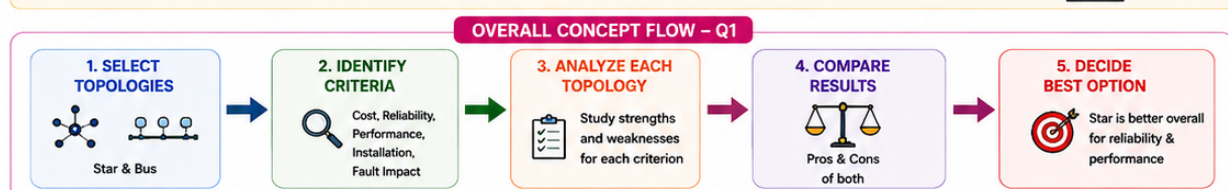
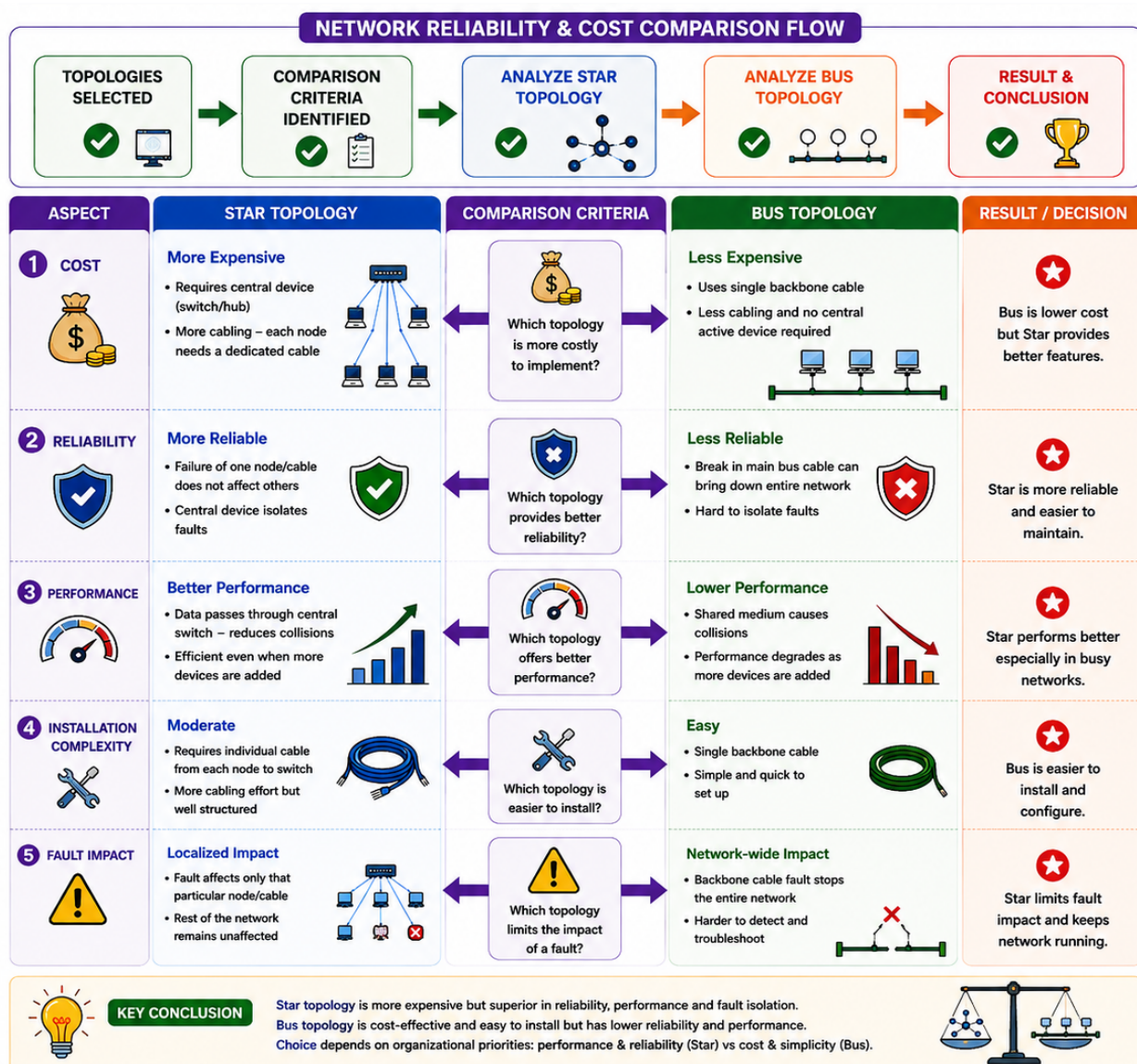
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QUESTION 1

Question: Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Comparison on Star and Bus topology in terms of cost, reliability and performance

CONCEPT MAP



Analysis

Star topology requires a central switch or hub and a separate cable for each node, so its initial cost and cabling requirement are higher. Bus topology uses one shared backbone, making it cheaper and simpler to install. However, star is more reliable because failure of one node or its cable normally does not affect the remaining nodes. In bus topology, a backbone failure can disrupt the whole network. Star also gives better performance because the switch controls forwarding and avoids the heavy contention of a shared medium. As traffic increases, bus performance decreases due to the shared channel.

Decision / Insight

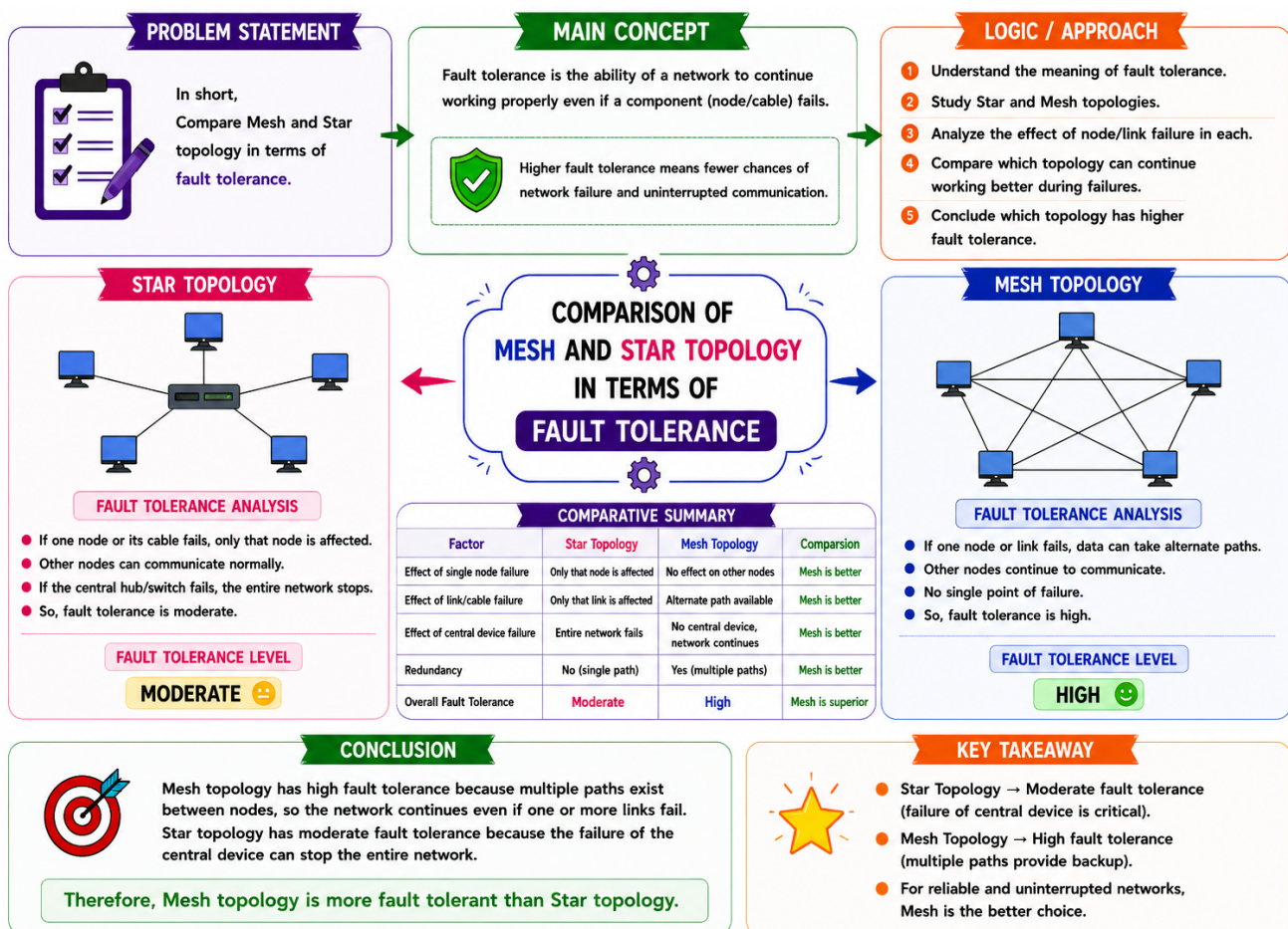
For a small, low-cost network, bus can be economical; for a reliable and higher-performing LAN, star is the better practical choice.

QUESTION 2

Question: Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Comparison on Mesh and Star topology in terms of fault tolerance

CONCEPT MAP



Analysis

Mesh topology provides very high fault tolerance because nodes have multiple communication paths. If one link fails, traffic can use an alternate path, so communication can continue. Star topology provides moderate fault tolerance: failure of one node link affects only that node, but failure of the central switch or hub can stop communication for all connected nodes. Therefore, mesh is more robust for critical networks where availability is more important than cost and complexity.

Decision / Insight

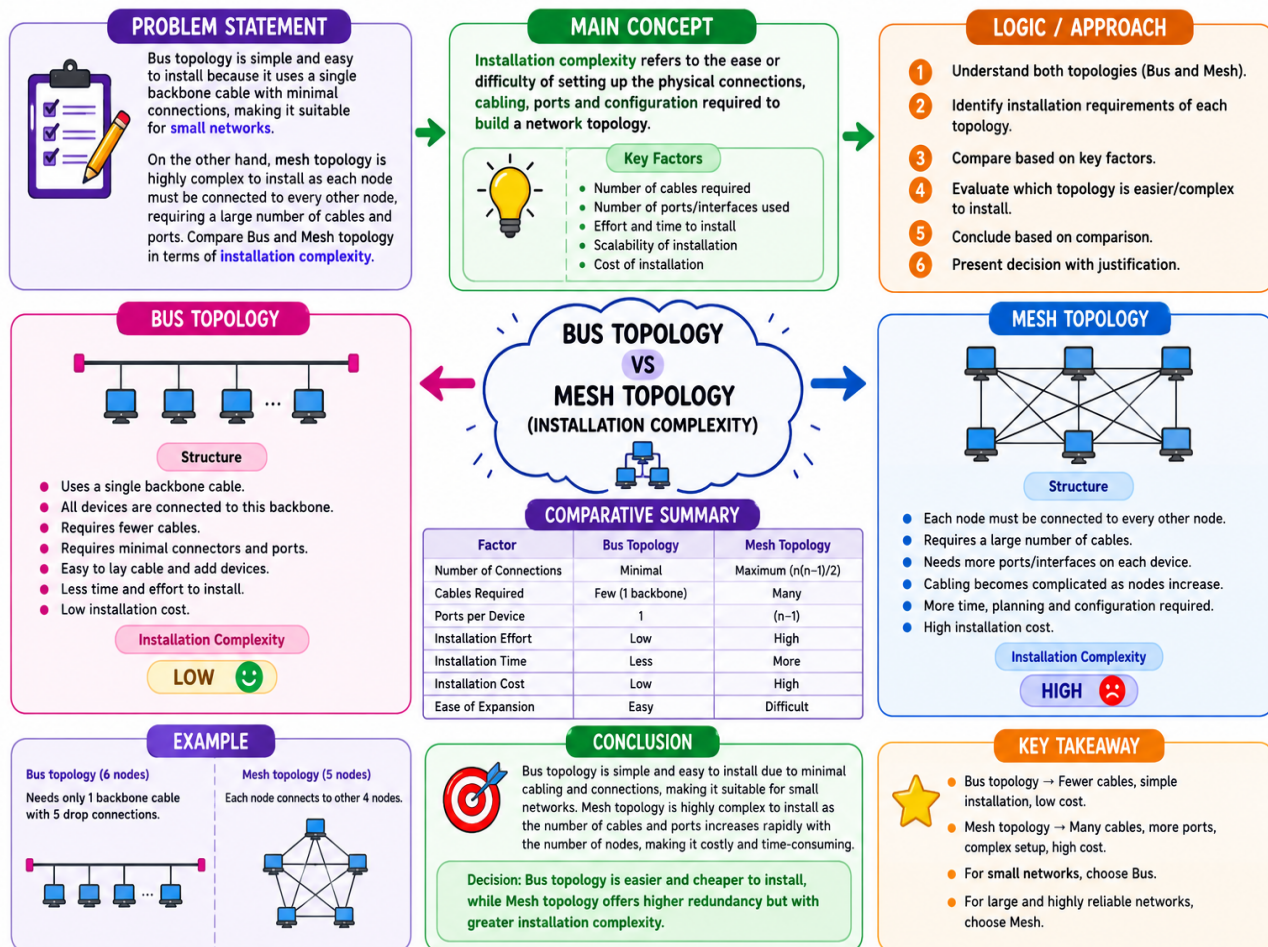
Mesh is preferred for critical and high-availability environments; star is preferred for normal LANs where simplicity and cost are important.

QUESTION 3

Question: Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Comparison on Bus and Mesh topology in terms of installation complexity

CONCEPT MAP



Analysis

Bus topology is simple because it uses a single backbone cable with relatively few connections. It requires less cabling and is easy to deploy for a small network. Mesh topology is much more complex because nodes need multiple direct links, increasing cable, port and configuration requirements. As the number of nodes grows, the number of links increases rapidly, making installation expensive and time-consuming.

Decision / Insight

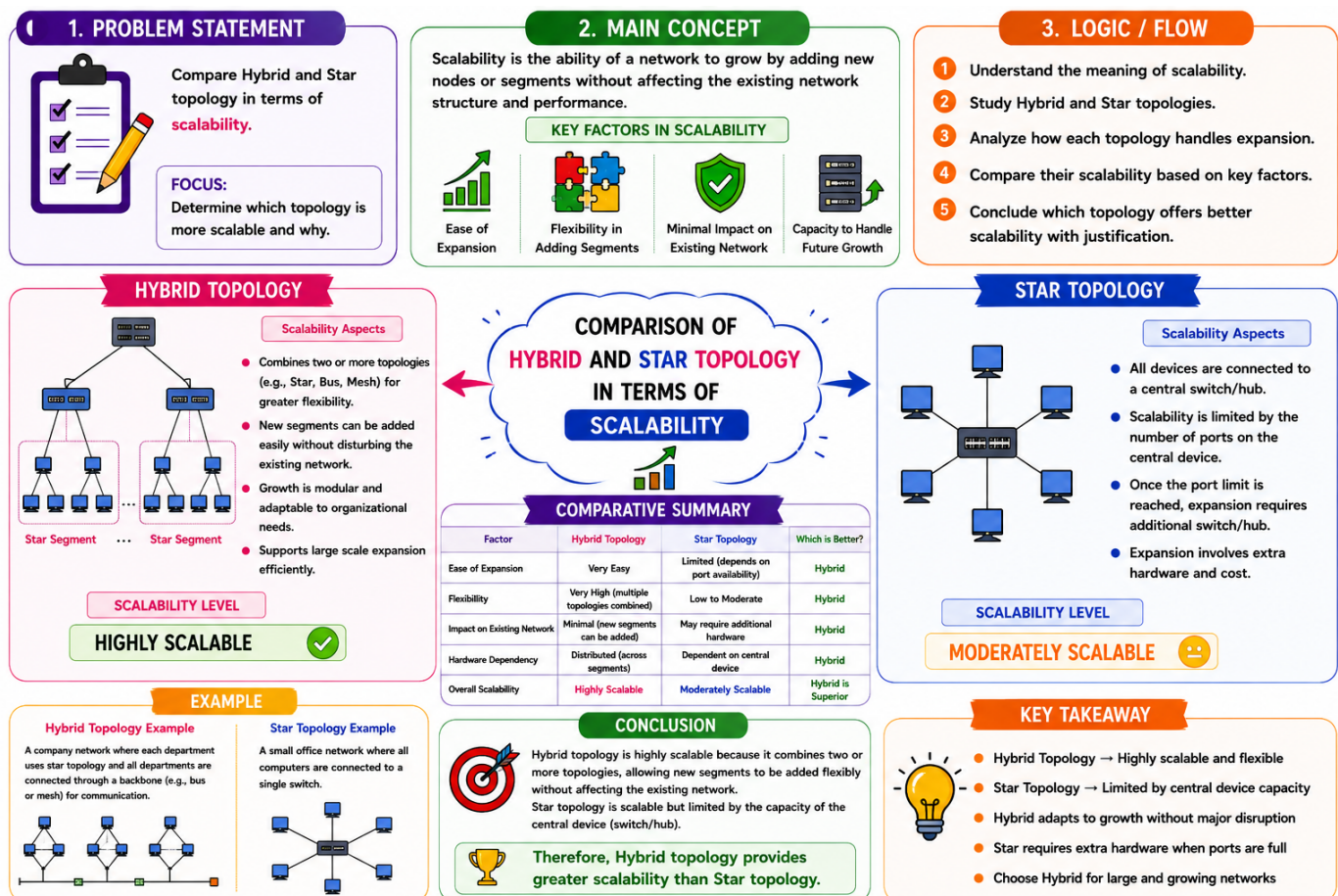
Bus is easier and cheaper to install, while mesh provides stronger redundancy at the cost of much greater installation complexity.

QUESTION 4

Question: Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Comparison on Hybrid and Star topology in terms of scalability

CONCEPT MAP



Analysis

Hybrid topology combines two or more topologies, such as star, bus or mesh. This modular structure allows new segments to be added according to organizational requirements, so it is highly scalable. Star topology is also scalable, but expansion depends on the available ports and capacity of its central switch or hub. When the central device reaches its limit, additional hardware is required. Hence hybrid topology offers greater flexibility for large and continuously expanding networks.

Decision / Insight

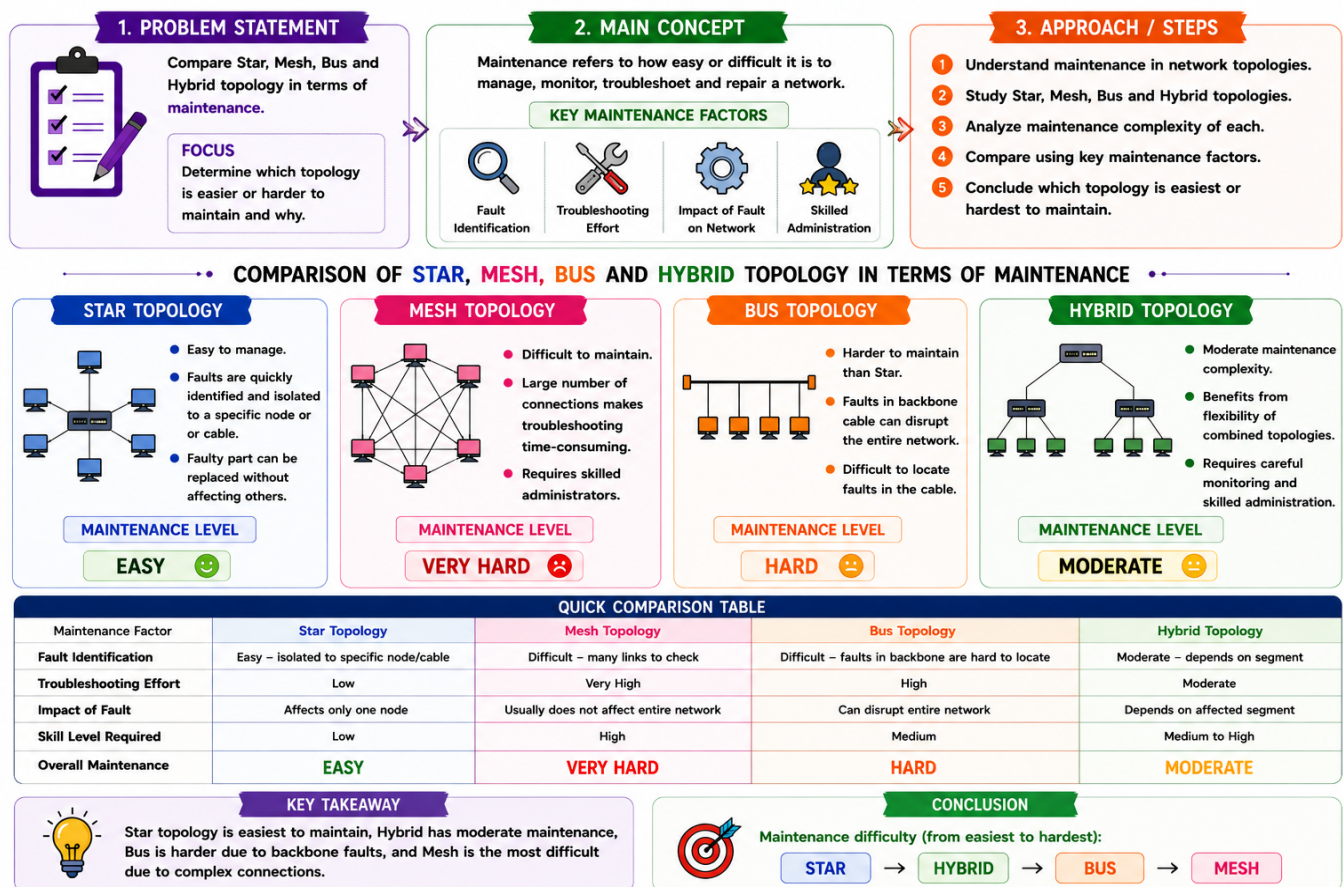
For a growing organization with different network requirements, hybrid topology provides better scalability; star is efficient for controlled LAN expansion.

QUESTION 5

Question: In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Comparison on Star, Mesh, Bus and Hybrid topology in terms of maintenance

CONCEPT MAP



Analysis

Star topology is the easiest to maintain because faults can usually be isolated to a particular node, cable or switch port. Mesh topology is highly reliable but difficult to maintain because its many links create more points to inspect and troubleshoot. Bus topology is harder to maintain because a backbone fault can affect the entire network and may be difficult to locate. Hybrid topology has moderate maintenance complexity because it combines multiple structures; its flexibility is useful, but administrators must understand and monitor each component topology.

Decision / Insight

Overall, star provides the best balance of easy maintenance and practical reliability. Mesh gives the highest redundancy but requires more skilled maintenance, while hybrid offers flexibility with moderate complexity.